LEARNING WHAT TO CHANGE

From Visual Reasoning to Scientific Hypothesis Revision

Chimdi Walter Ndubuisi · University of Missouri REVIEW DRAFT 19 October: for collaborator review, not for print

Can a reasoner recognize when its explanation is inadequate, and determine what needs to change?

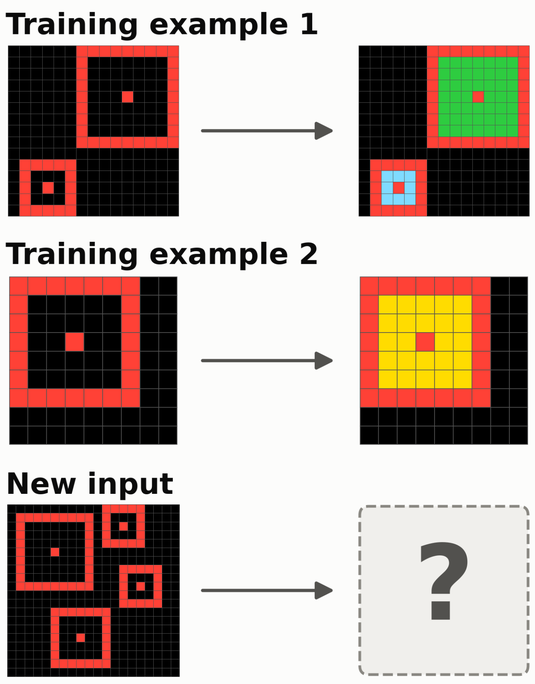
Solving a problem is not the same as knowing why the solution works, or what to change when the available concepts and rules fall short. A question shaped in discussions with Toni Kazic.

Aims, not yet all achieved: find objects and relations → write executable hypotheses → notice inadequacy → choose a justified revision → test it independently → transfer the procedure.

1 · Why ARC is a useful test

ARC tasks require inferring a transformation from a few examples and applying it to a new input.

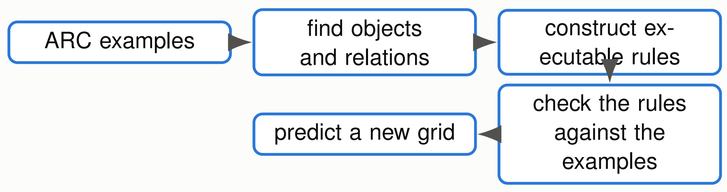
REAL ARC-AGI-2 TRAINING TASK 00dbd492 (public)



Each frame's fill colour depends on its size.

2 · CORA: what I built

CORA is an executable ARC reasoning system used to investigate how models construct, test and revise hypotheses from sparse visual examples.



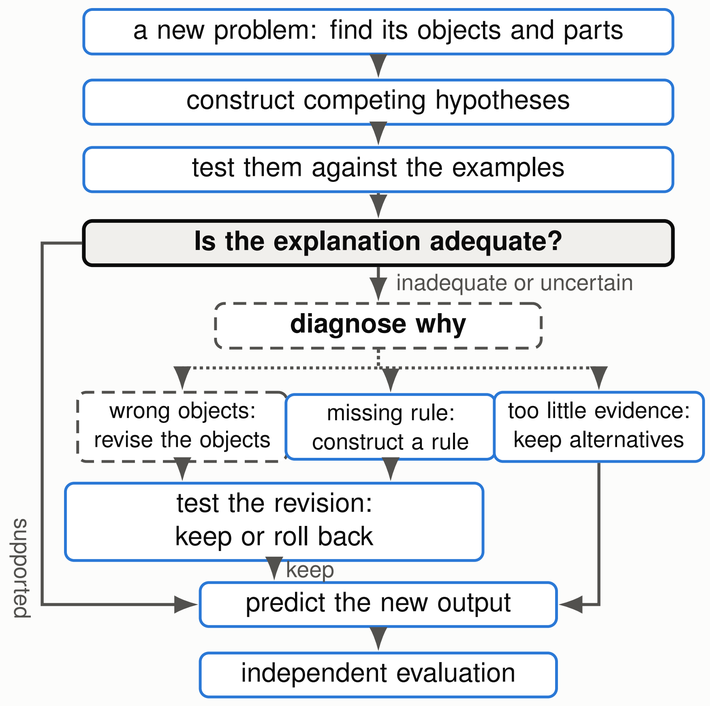
Fitting is not enough [DONE]

6
answers accepted, all correct

26
tasks where every rule that fit the examples failed the test

60 exposed ARC-AGI-2 training tasks; CORA's object reasoner as submitted.

3 · Learning what to revise



Recognizing an inadequate explanation: partly built (CORA-v2). **Knowing which revision helps**: not yet shown. CORA never sees the hidden answer.

Four reasons to revise

ILLUSTRATIVE — drawn to explain, not a CORA result

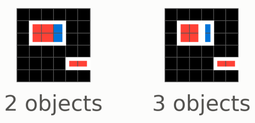
A Memorization

A rule that copies the example fits it, then fails a new input.



B Wrong objects

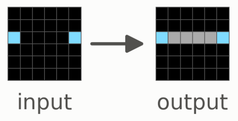
The same grid can be split into objects in more than one way.



A rule written for one split can fail on the other.

C Missing rule

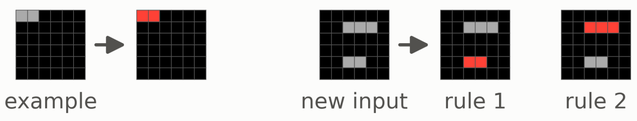
The right objects, but no operation describes the change.



operations: move, recolour
✗ neither can draw the line

D Not enough evidence

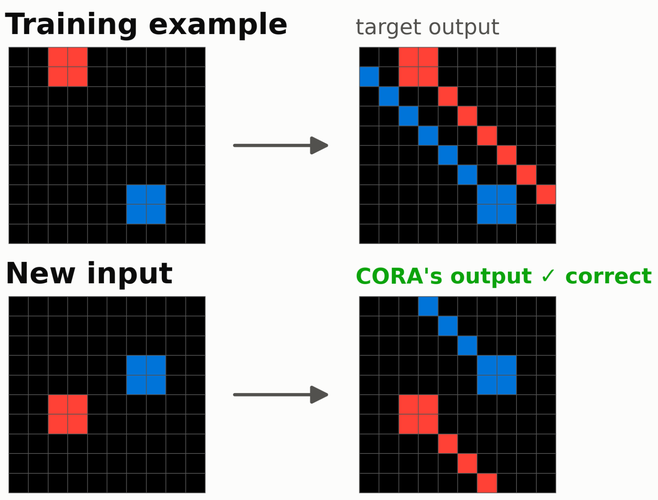
Rule 1: bars of 2 turn red. Rule 2: top bars turn red.
Both fit the example; they disagree on a new input.



These are different possible causes. Telling them apart, from the examples alone, is the research problem.

A rule CORA built [DONE, NARROW]

REAL ARC-AGI-2 TRAINING TASK ka95aa62f67d9 · recorded run



Depending on an object's colour, extend a line toward one of two corners: **blue** up-left, **red** down-right.

Automatically constructed conditional operation; rebuilt 3 of 3 times with one example held out. An exposed task chosen in development; a constructor that ignored the failure record found the same rule, so diagnosis did not cause this gain.

Did it generalize? [NEGATIVE]

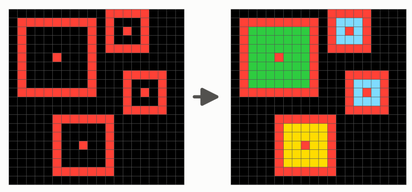
0 / 58
new verified solves from rule construction

58 separately selected, still exposed training tasks; every construction variant; no wrong answers; no advantage from the failure record.

CORA-v2 [DEVELOPMENT]

Compares alternative executable hypotheses and records why it is uncertain.

REAL TASK 00dbd492 · CORA-v2 development run
CORRECT DEVELOPMENT PREDICTION — NOT CERTIFIED



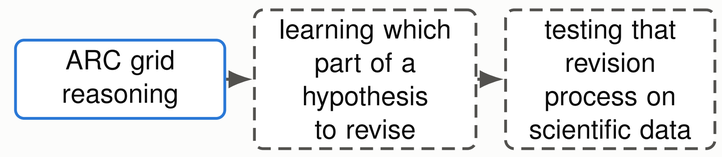
fill colour, by size:
azure: 3 examples
yellow: 2 examples
green: 1 example
hide that example:
the check fails

On 118 exposed development tasks: 4 correct and 1 wrong exploratory guesses; nothing certified.

Inventing new operations [RUNNING]

Automatic rule-language extension, a separate CORA experiment: can a system build new reusable operations itself? 475 of 497 proposal units done (19 October); no verdict; not part of CORA-v2 or any Kaggle entry.

From ARC to science [FUTURE, NOT TESTED]



Proteins · single-cell and omics data · plant phenotypes · pathways suggested by maize lesion mimics, checked with molecular or perturbation evidence. What could transfer is the revision policy, not ARC operations.

Building a rule is not the same as learning which revision improves an unseen prediction. That is what we are trying to test.

implemented, tested · proposed, not yet validated

[DONE] [NEGATIVE] [DEVELOPMENT] [RUNNING] [FUTURE, NOT TESTED]

Two approaches to ARC, official public ARC-AGI-2 scores

NVARC / Qwen3-4B: 27.64%	**CORA: 0.00%** (also 0.00% with rule construction)
reproduced third-party method, **not CORA**; neural training and per-task adaptation	original research; explicit hypotheses, tested revisions

Open question: can learned neural proposals help an executable reasoner find which part of a hypothesis to revise?

Related work: Popper (Cropper & Morel 2021); DreamCoder (Ellis et al. 2021); LILO (Grand et al. 2024); Model Discovery Agent (Murphy 2026). The failure-to-revision idea is not new; the question is which component to revise, against matched controls. Numbers: `records/CORA_SUMMIT_CLAIM_LEDGER_20261017.md` · AI coding assistance used for parts of the code and text · authorship pending collaborator review.